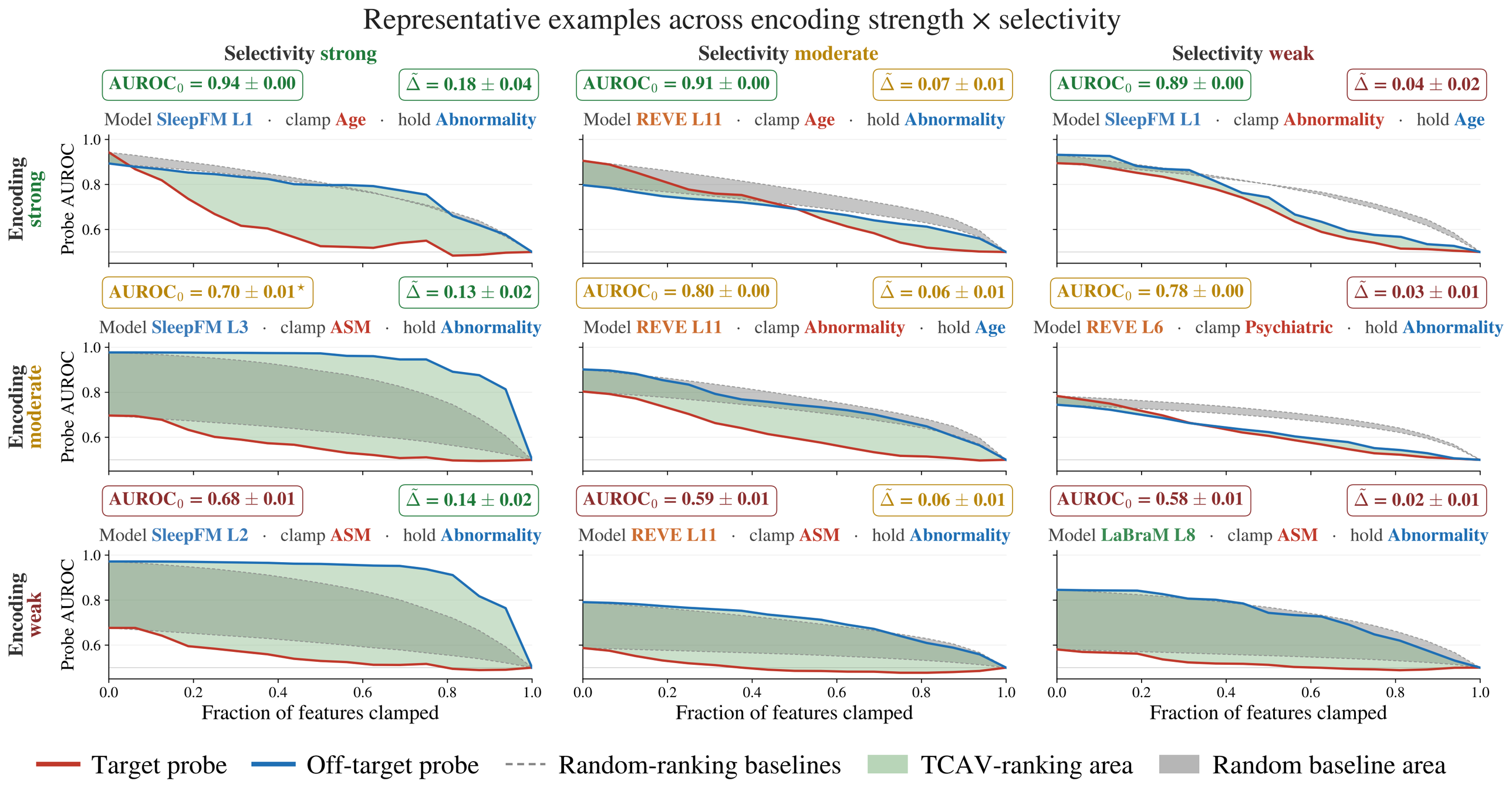
Representative examples across encoding strength × selectivity
Selectivity strong
Selectivity moderate
Selectivity weak
Encoding strong
Encoding moderate
Encoding weak
AUROC$_0$ = 0.94 ± 0.00
Δ̃ = 0.18 ± 0.04
Model SleepFM L1 · clamp Age · hold Abnormality
AUROC$_0$ = 0.91 ± 0.00
Δ̃ = 0.07 ± 0.01
Model REVE L11 · clamp Age · hold Abnormality
AUROC$_0$ = 0.89 ± 0.00
Δ̃ = 0.04 ± 0.02
Model SleepFM L1 · clamp Abnormality · hold Age
AUROC$_0$ = 0.70 ± 0.01*
Δ̃ = 0.13 ± 0.02
Model SleepFM L3 · clamp ASM · hold Abnormality
AUROC$_0$ = 0.80 ± 0.00
Δ̃ = 0.06 ± 0.01
Model REVE L11 · clamp Abnormality · hold Age
AUROC$_0$ = 0.78 ± 0.00
Δ̃ = 0.03 ± 0.01
Model REVE L6 · clamp Psychiatric · hold Abnormality
AUROC$_0$ = 0.68 ± 0.01
Δ̃ = 0.14 ± 0.02
Model SleepFM L2 · clamp ASM · hold Abnormality
AUROC$_0$ = 0.59 ± 0.01
Δ̃ = 0.06 ± 0.01
Model REVE L11 · clamp ASM · hold Abnormality
AUROC$_0$ = 0.58 ± 0.01
Δ̃ = 0.02 ± 0.01
Model LaBraM L8 · clamp ASM · hold Abnormality
Probe AUROC
1.0
0.8
0.6
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of features clamped
Target probe
Off-target probe
Random-ranking baselines
TCAV-ranking area
Random baseline area